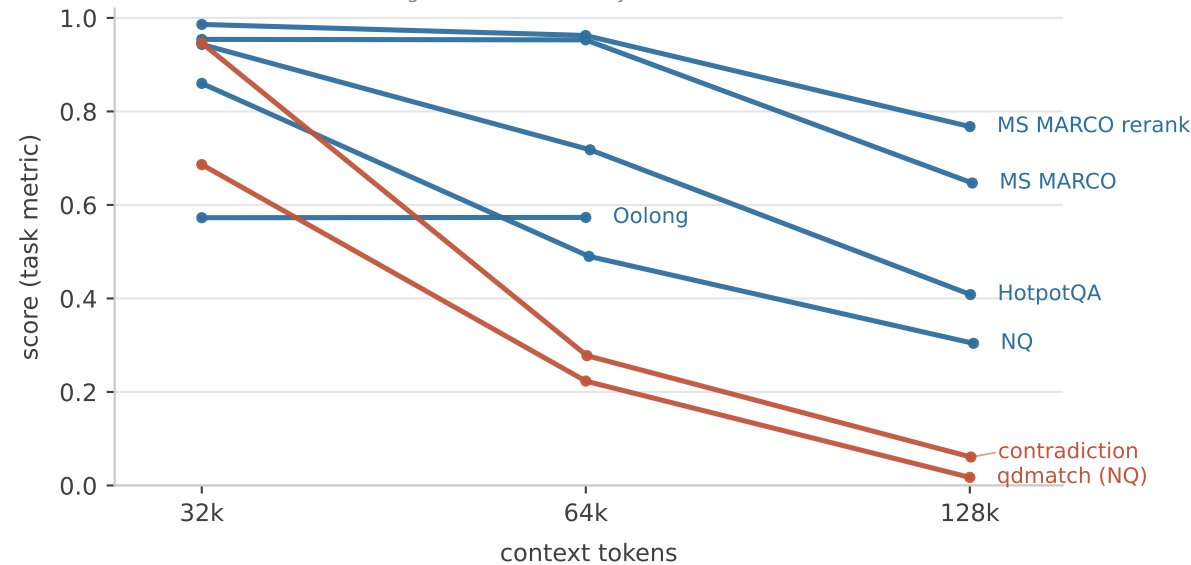


Past the 32k training ceiling, low-CTC tasks decay and high-CTC tasks fall off a cliff

full (dense) attention, Qwen3.5-4B, evaluated on 64k and 128k rungs of tasks it was only ever trained on up to 32k

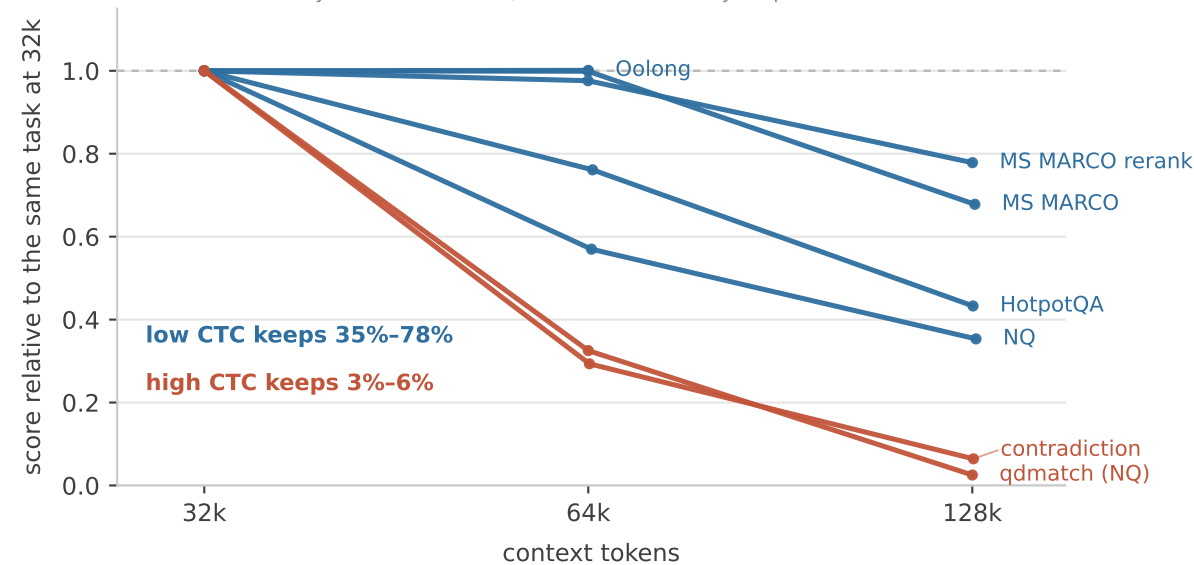
(a) absolute score on the long rungs

7 tasks with two or more rungs · full attention only



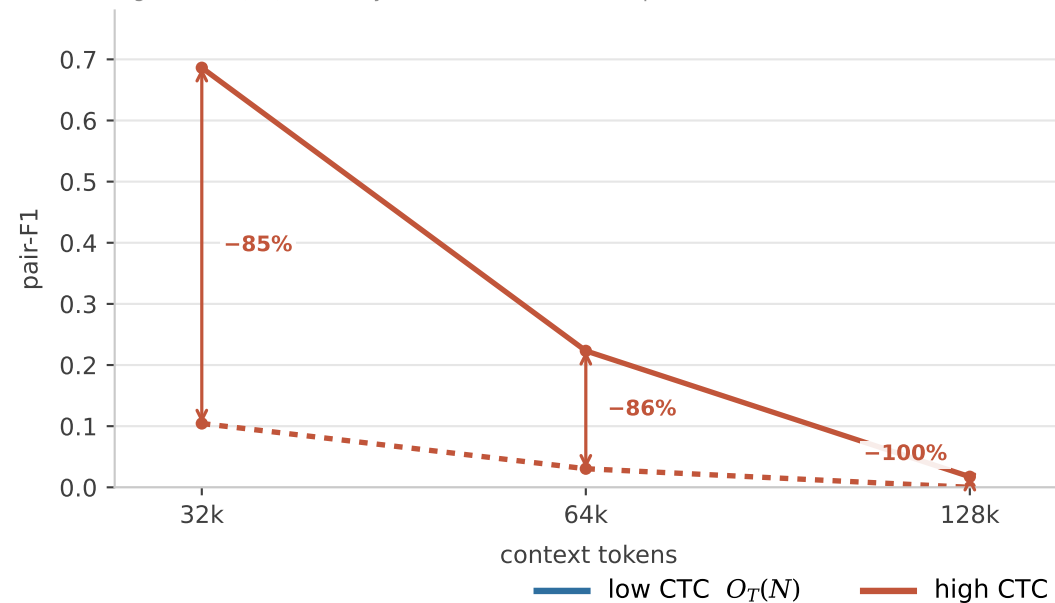
(b) retention past the ceiling

each task divided by its own 32k score, so absolute difficulty drops out



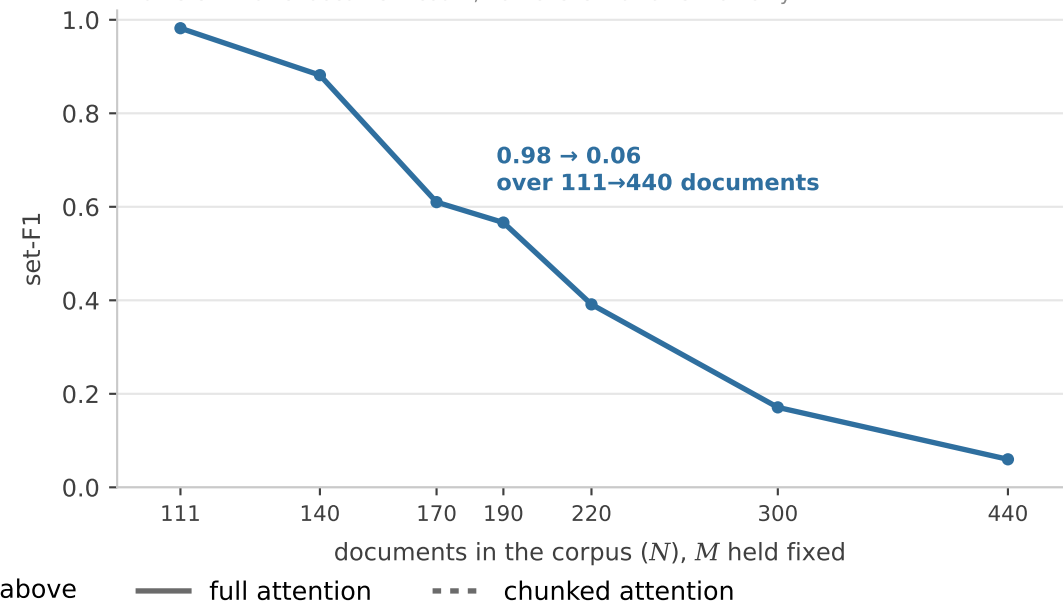
(c) qdmatch (NQ): the chunked arm dies first

high CTC $O_T(N^2)$ · the only task with a chunked run past 32k



(d) outlier (wiki, fixed M): collapse tracks N

△ different x-axis: document count, not tokens · full attention only



Excluded long rungs (14 of 45 exported rows):

- absence — RETRACTED: gold_semantics=absence, so injected filler documents silently satisfy the gold condition and become unlabelled true positives. Needs...
- cycle, outlier (Amazon), outlier (wiki, scale-k), textgroups — RETRACTED: gold_semantics=structural, so injected filler documents silently satisfy the gold condition and become unlabelled true positives. Needs...
- NIAH-contra — two independent 64k/128k niah builds disagree (0.736 vs 0.002 at 64k); ladder needs a rebuild before any of it is quoted
- NIAH-contra, outlier (Amazon), outlier (wiki, scale-k), textgroups — 32k rung only once the exclusions above are applied, so no length-generalisation slope can be drawn